



# Real-Time Chess Tracking System

---

A Computer Vision Project for Chess  
Game Detection and Validation

Panagiotis Kounelis, Konstantinos Megaritis, Ioannis  
Zouras, Ifigenia Mavraki, Shalva Bakradze



# Project Overview

---

- A system to detect and track a real chess game using computer vision techniques.
- It processes a chessboard in real-time via a webcam feed.
- The system identifies moves, validates them according to chess rules, and visualizes updates.
- Phases include chessboard calibration, piece initialization, move tracking, and game conclusion.

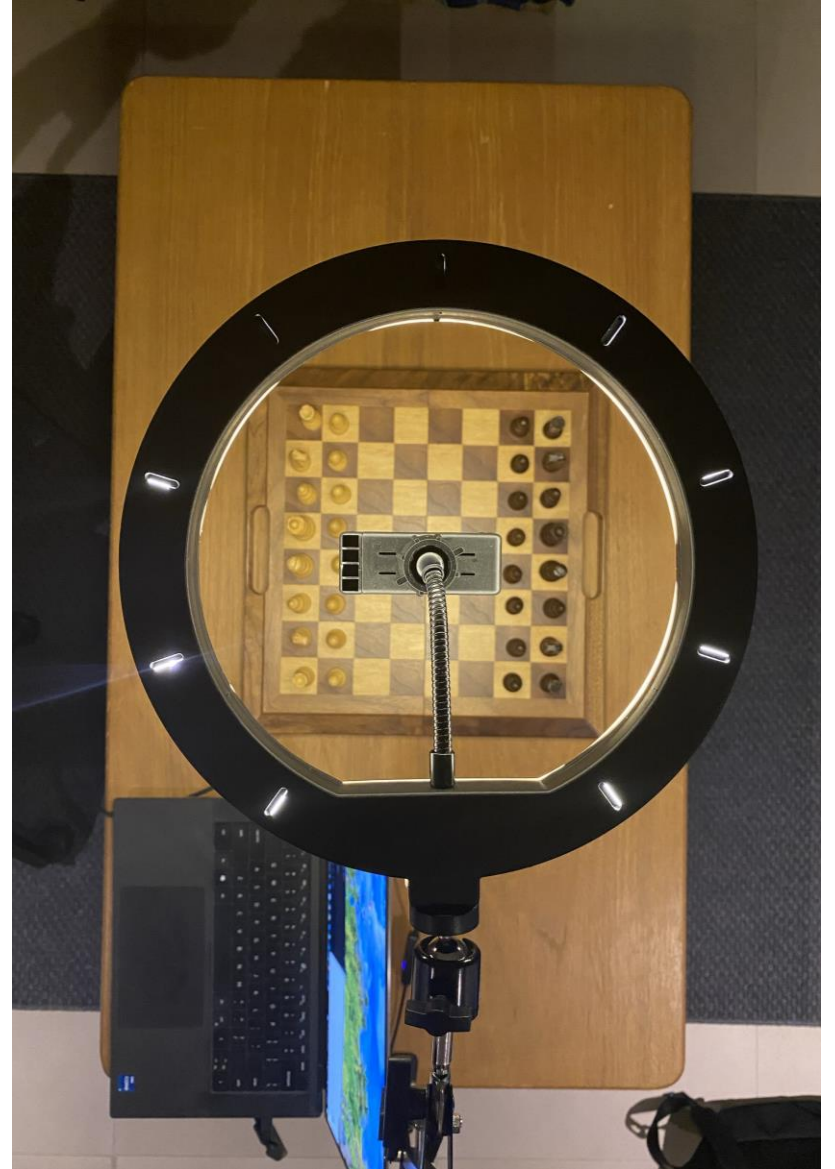


# Assumptions

---

- The camera is positioned directly above the chessboard.
- The camera is placed to the right side of the player controlling the white pieces.
- The camera's angles are perfectly aligned with the corners of the chessboard.
- The gray value of the background is not equal to the gray value of the chessboard.





# Key Libraries and Tools

- OpenCV: Video – Image processing (Canny edge detection, dilation, erosion, connected components, transformation, subtraction).
- NumPy: Matrix operations and logical manipulation.
- copy: For creating deep copies of arrays
- Counter: For comparing arrays
- os: File handling for saving game states.
- Python Standard Libraries: Logical implementation for validation and game logic.

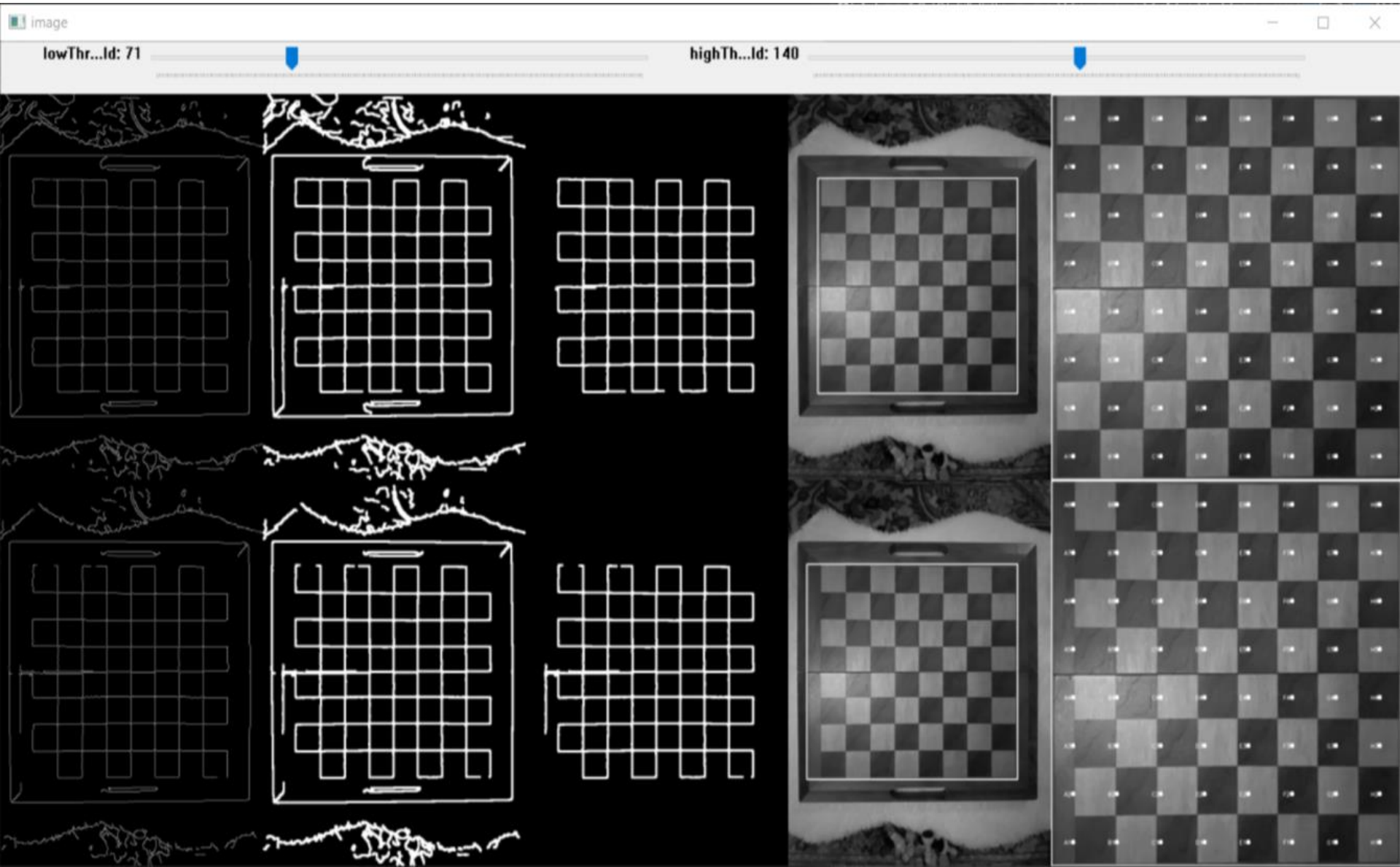


# Phase 1: Chessboard Calibration

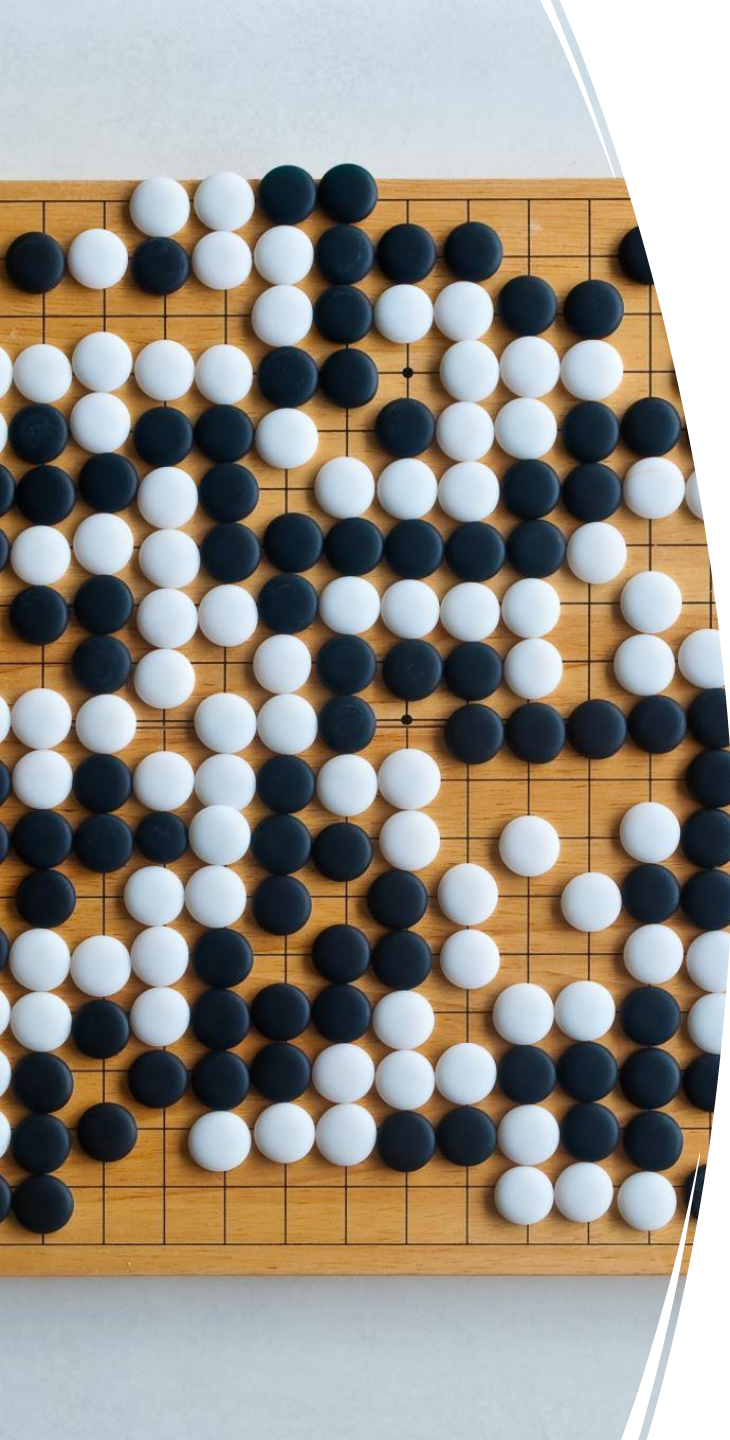
---

- Median Blurring – Gaussian Blurring
- Use Canny edge detection to detect edges of the physical chessboard.
- Dilation.
- Connected Component Analysis with Stats.
- Dynamic threshold trackbars to fine-tune edge detection.
- Apply perspective transformation to align and normalize the chessboard to an 800x800 grid.
- Ensures accurate positioning for move detection.

# Threshold Calibration





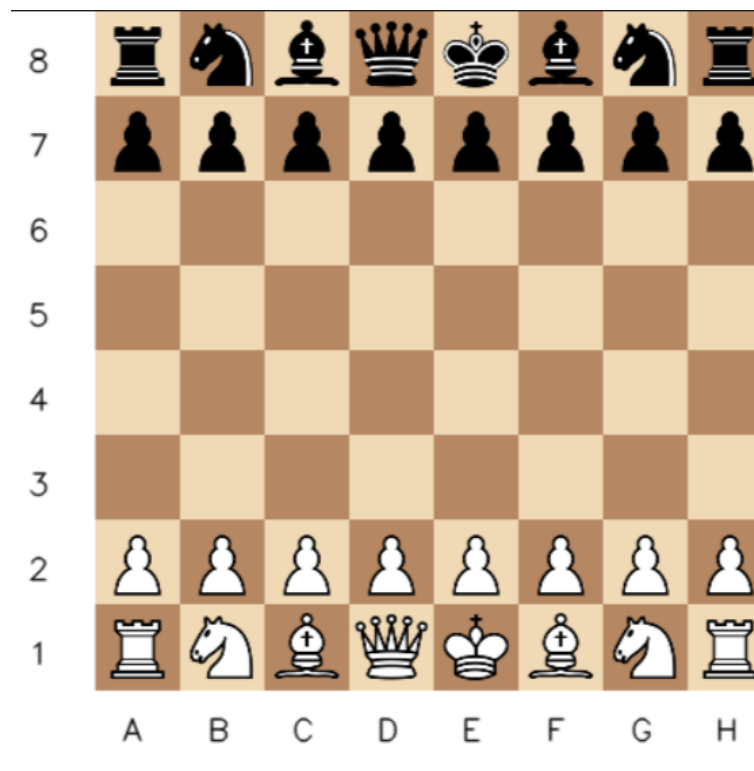
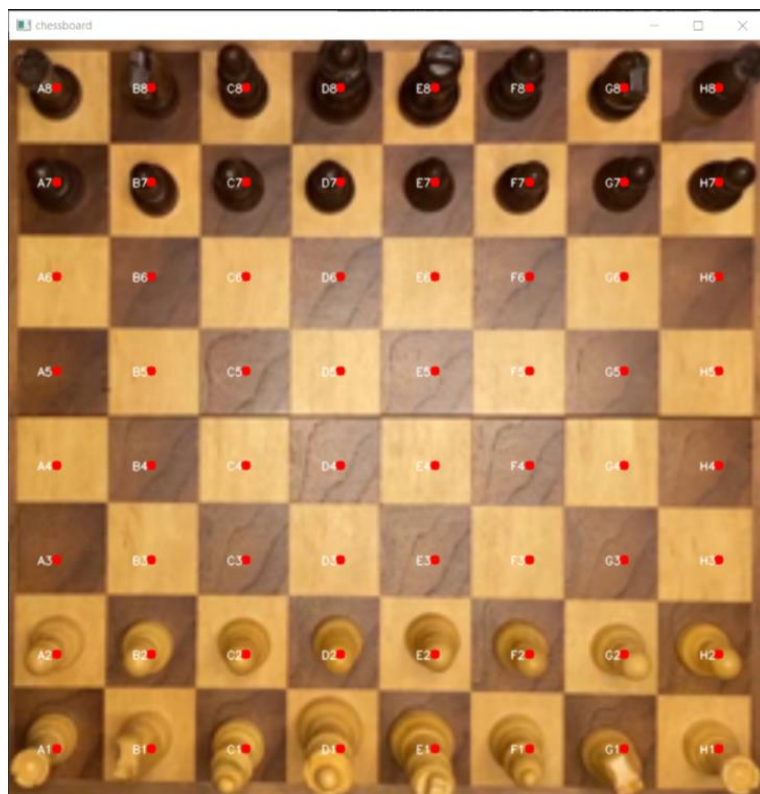


# Phase 2: Piece Initialization

---

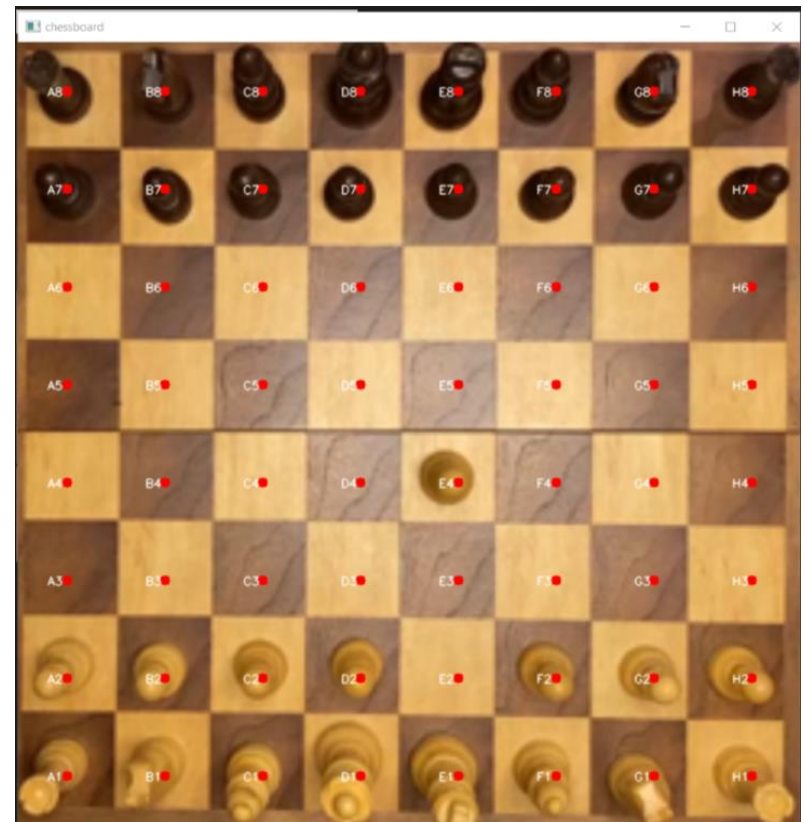
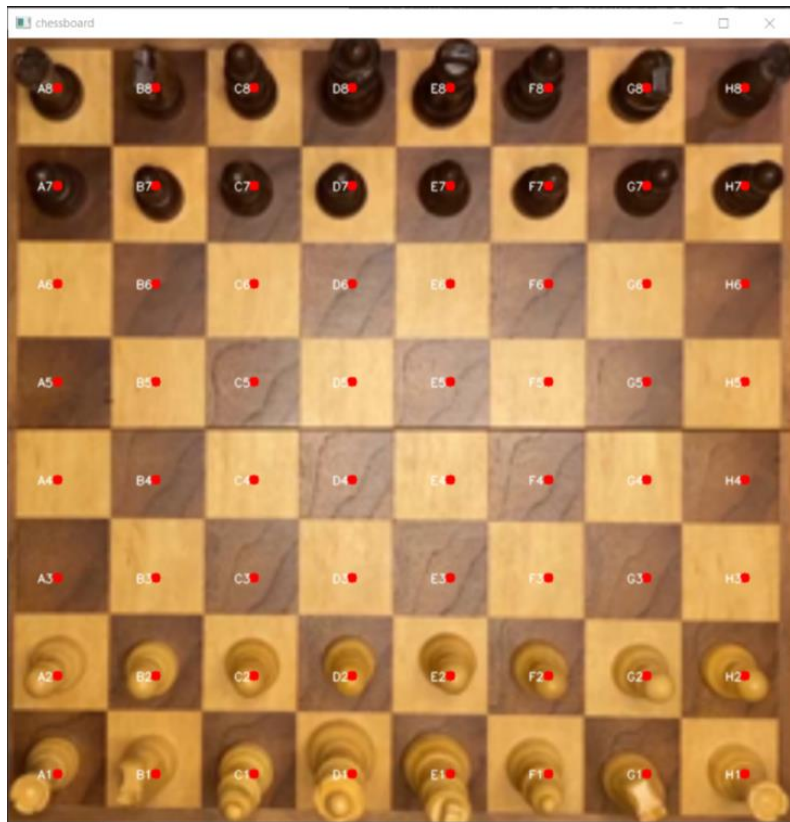
- Once the calibration is completed, neither the camera nor the chessboard should move.
- Map each square on the chessboard grid to its corresponding real-world coordinates.
- Centers of all squares are calculated and annotated for alignment.
- Initial chess piece positions are placed on the physical chessboard.
- Virtual chessboard matrix acts as a reference point for move tracking.





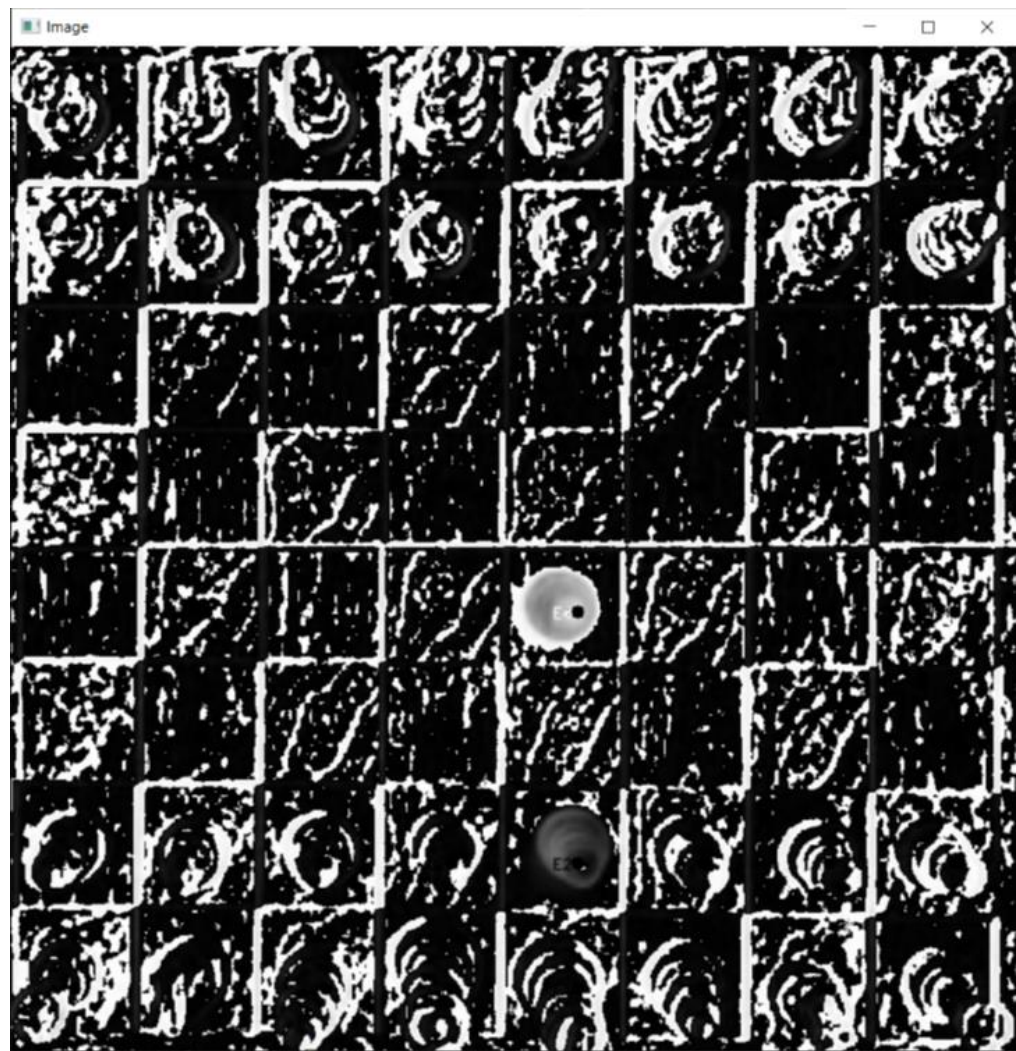
# Phase 3: Game in Progress

- Capture a snapshot of the board after each move.
- Use image subtraction to detect changes between previous and current snapshots.
- Band thresholding to create a binary (B&W) image, highlighting significant changes.
- Erosion to further clean noise, followed by more rounds of dilation to emphasize the detected movements.



# Image subtraction

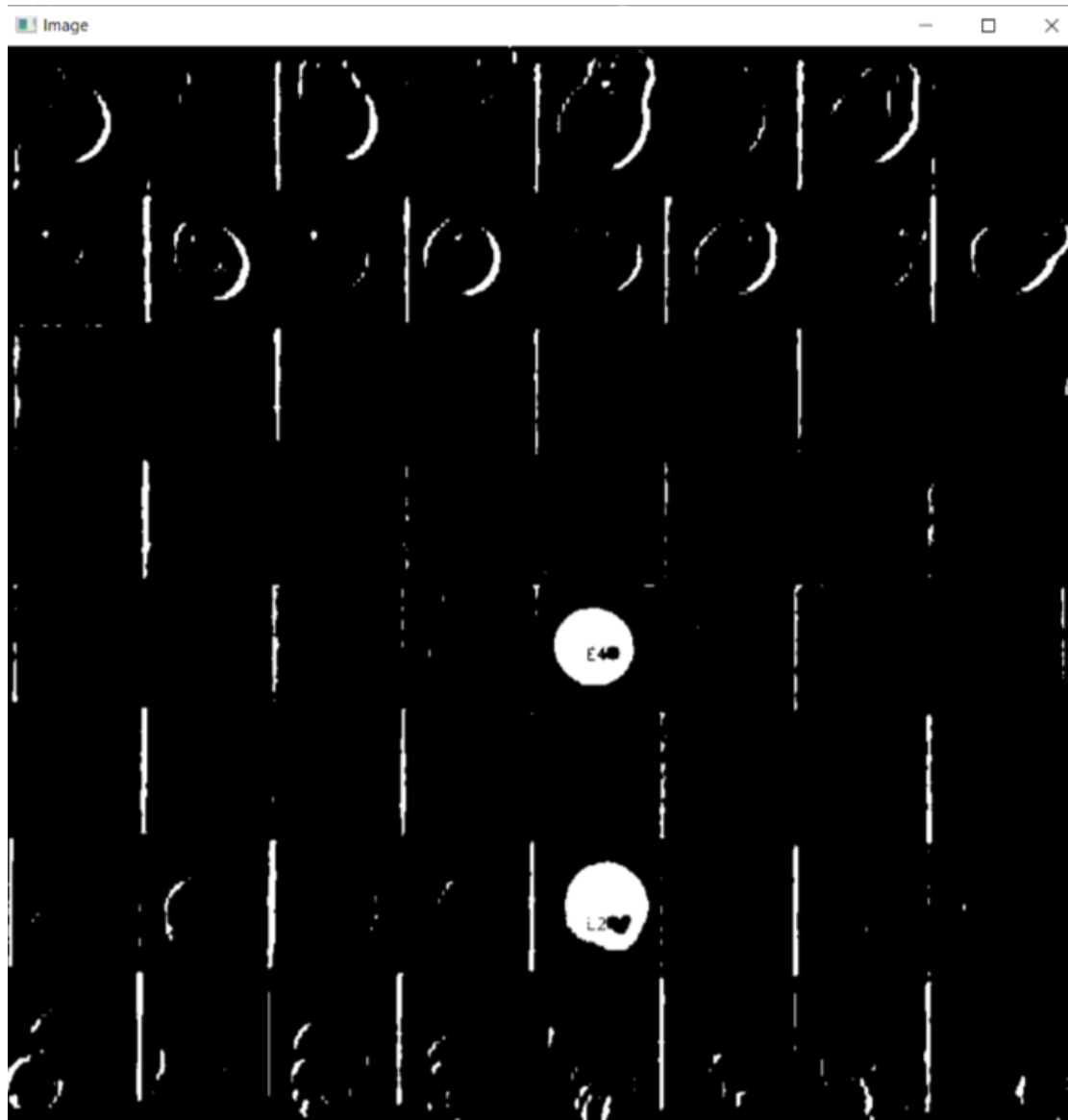
---





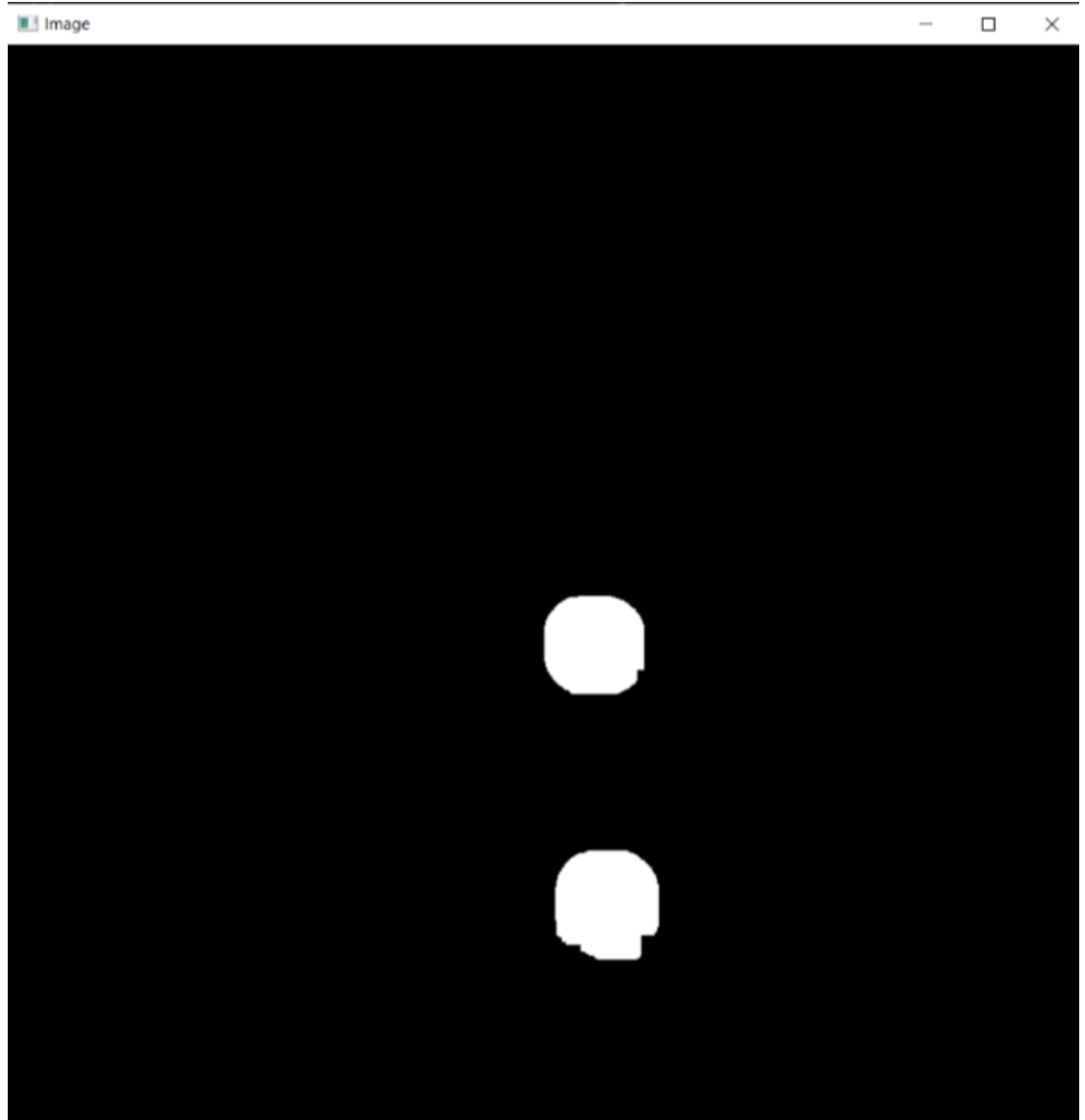
# Band thresholding

---



# Erosion- Dilation

---



# Connected Components Analysis

- Identify the two largest non-black components in the processed image.
- These correspond to the 'start' and 'end' positions of the moved piece.
- Logical mapping connects changes to the virtual chessboard matrix.
- Move validation checks ensure rules are adhered to.

# Move Validation

---

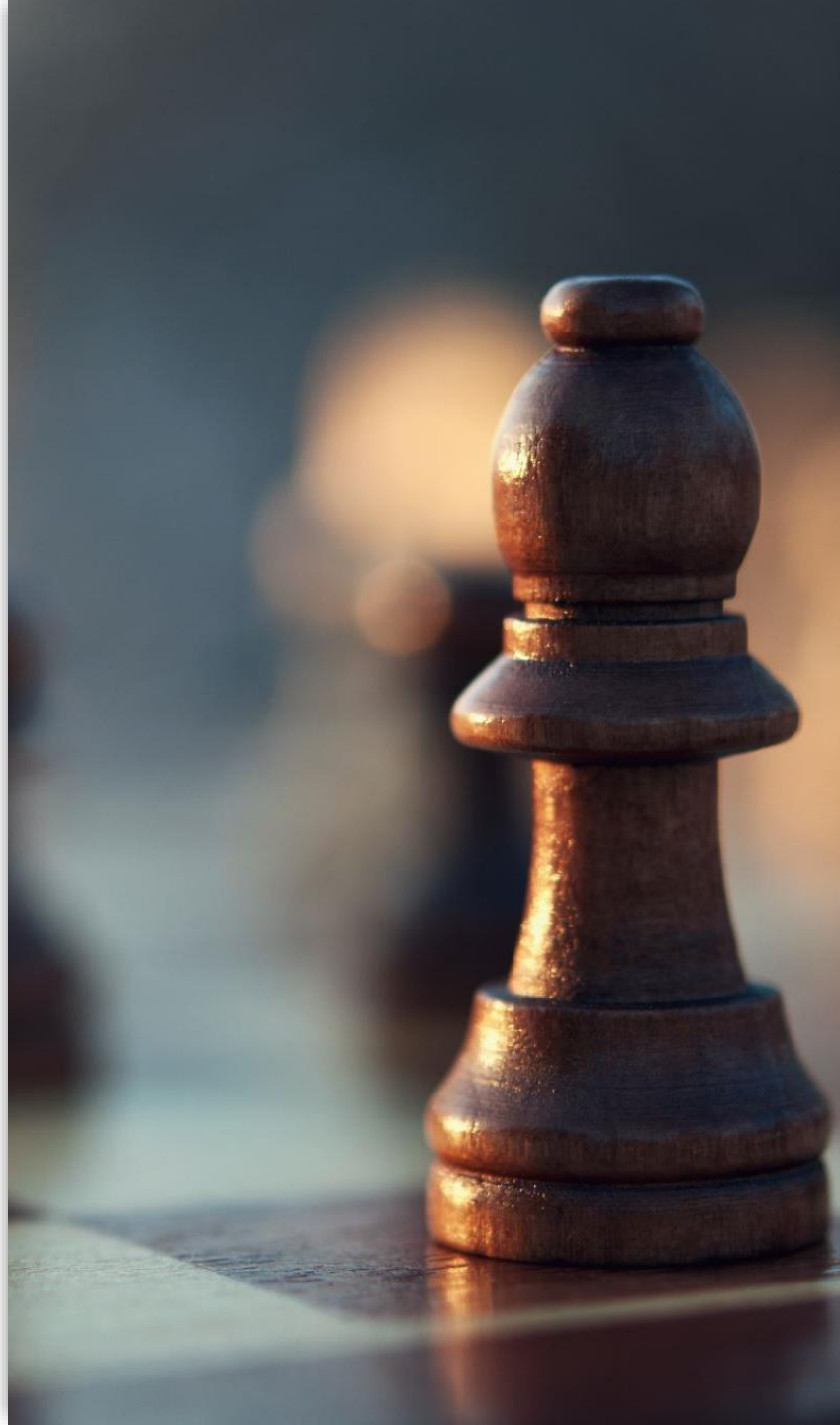
- Validates moves for each piece (Pawn, Rook, Knight, Bishop, Queen, King).
- Prevents invalid moves and ensures compliance with chess rules.
- Special checks for:
  - Pawn movement (initial 2-step move, diagonal capture)
  - Rook and Bishop line of sight
  - Knight's L-shaped move
  - Castling, En Passant, Pawn Promotion
- Checks for illegal states like placing own King in check.





# Checkmate Detection

- Identify the opponent's King position.
- Check if the King is in check.
- Evaluate if the opponent has any legal moves to resolve the check.
- If no legal moves exist while in check, it is checkmate.



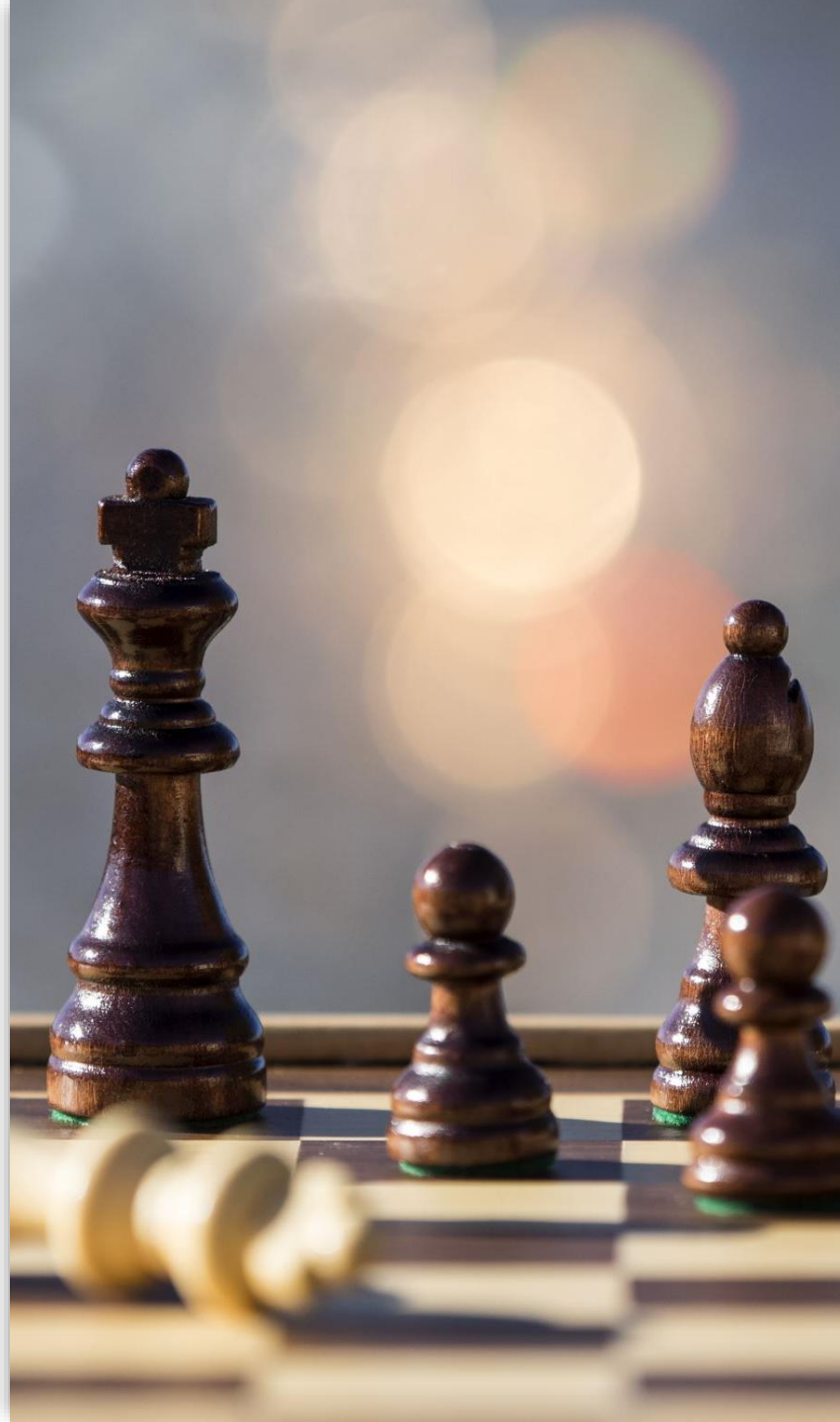
# End Game State

The game has 4 possible endings:

- Checkmate, the game concludes with a winner.
- Draw after agreement (Pressing D)
- Resignation (Pressing Q)
- App Crash 🤡

Post Game Actions:

- Game states (chessboard 2D images) can be saved for analysis (Pressing S)
- Players can also simply exit the program manually (Pressing Esc)

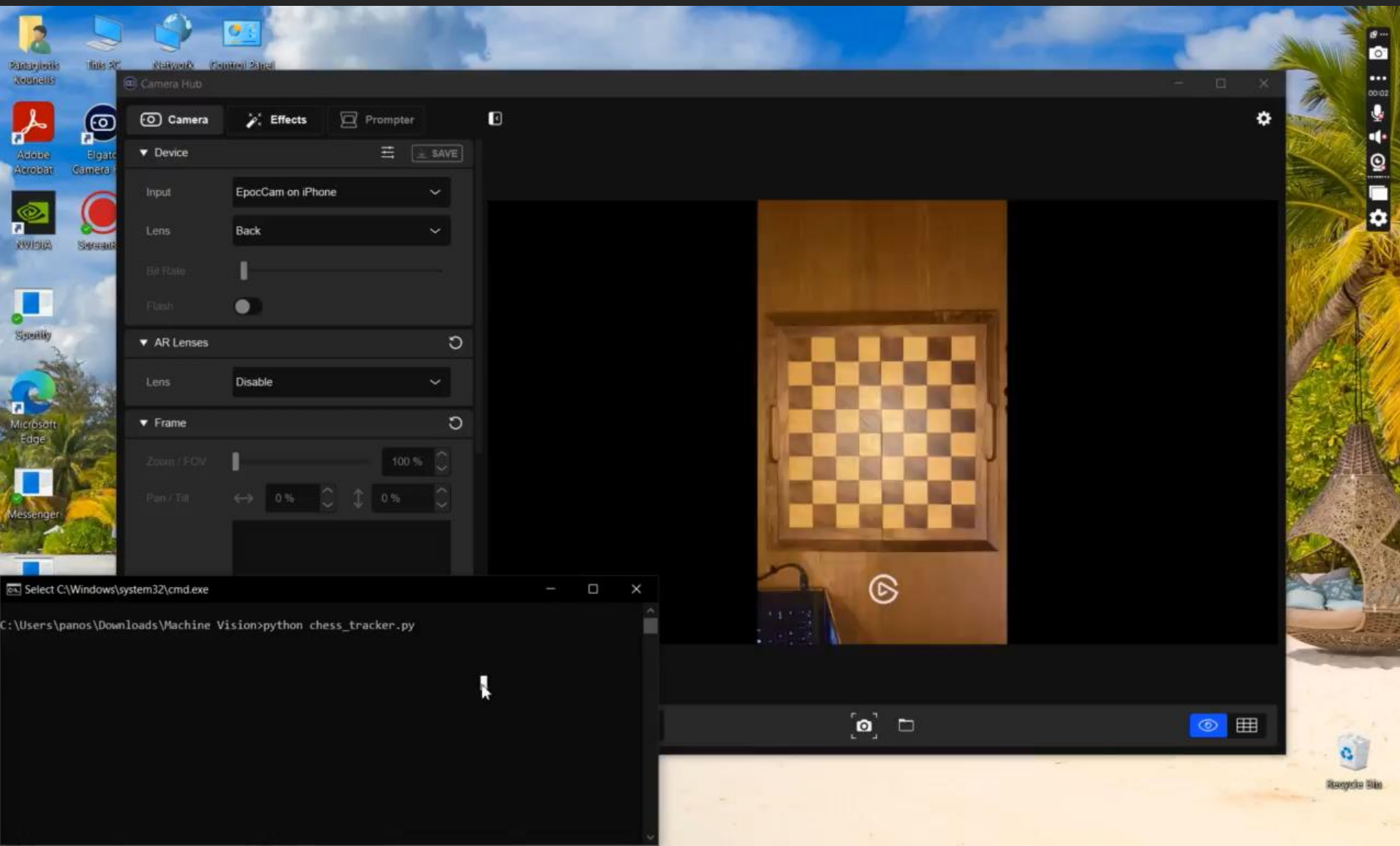


# Conclusion

- The system successfully implements a real-time chess tracking solution.
- Combines computer vision with game logic to monitor and validate chess games.
- Potential for extensions:
  - Multi-angle camera support.
  - Incorporate clock functionality.
  - Inclusion of other end game conditions like draw by stalemate, fifty-move rule and threefold repetition.
  - Integration with AI for piece recognition and game analysis.
  - Expansion to other board games.

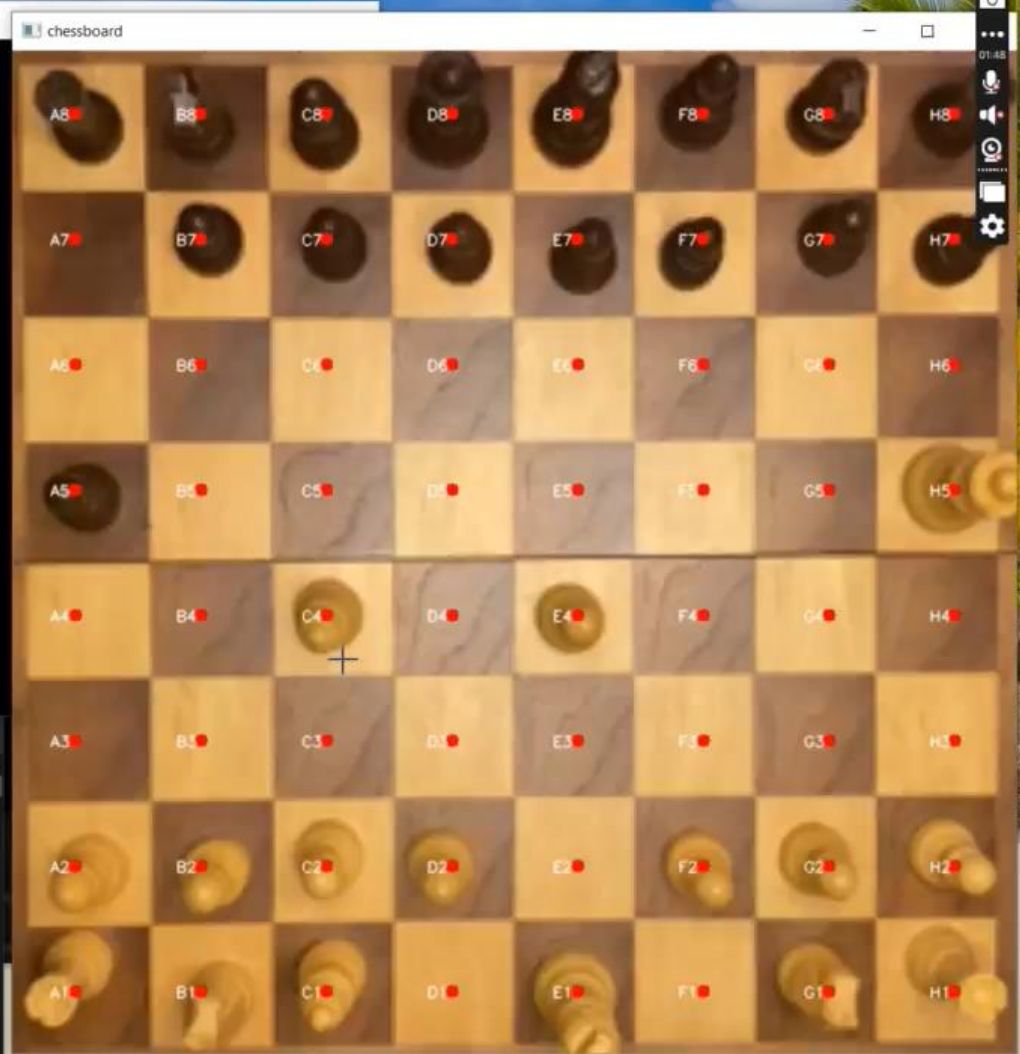
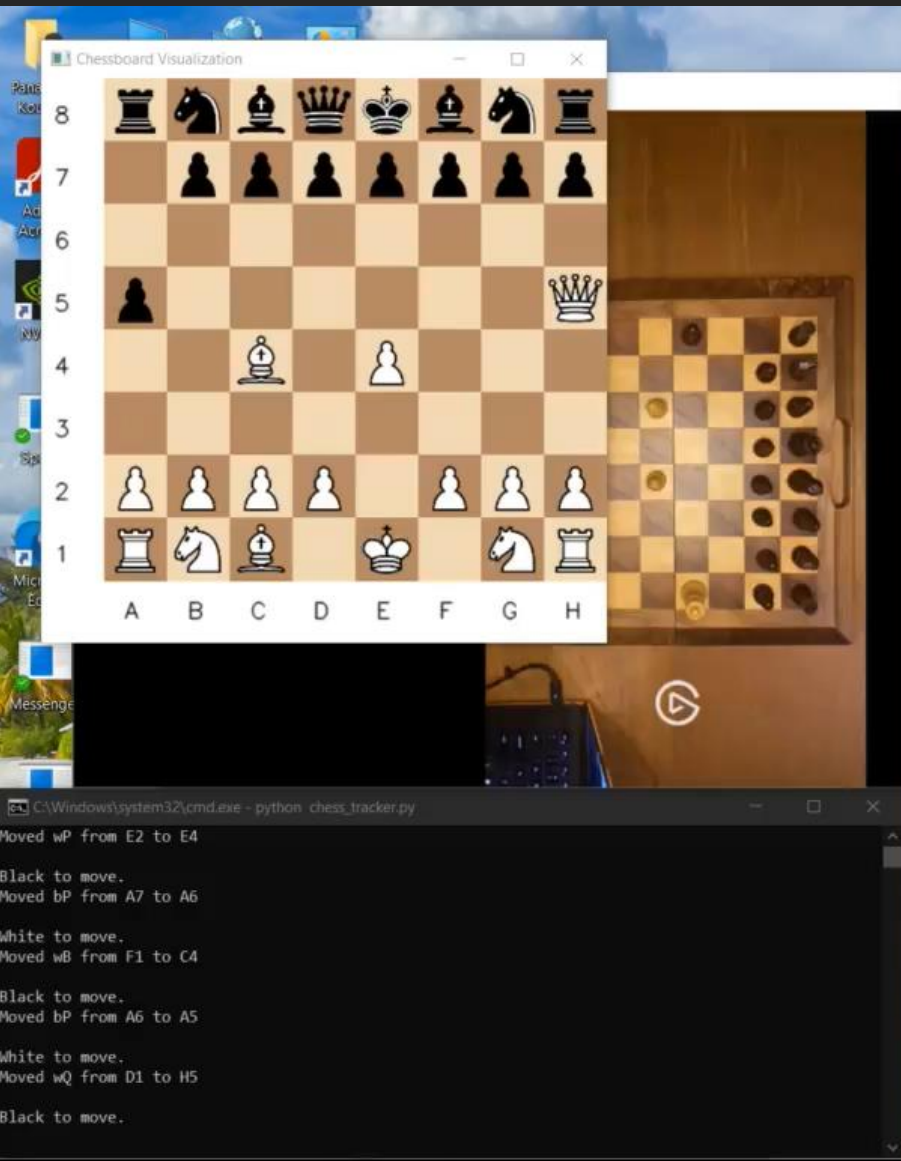


# Demo Tracker



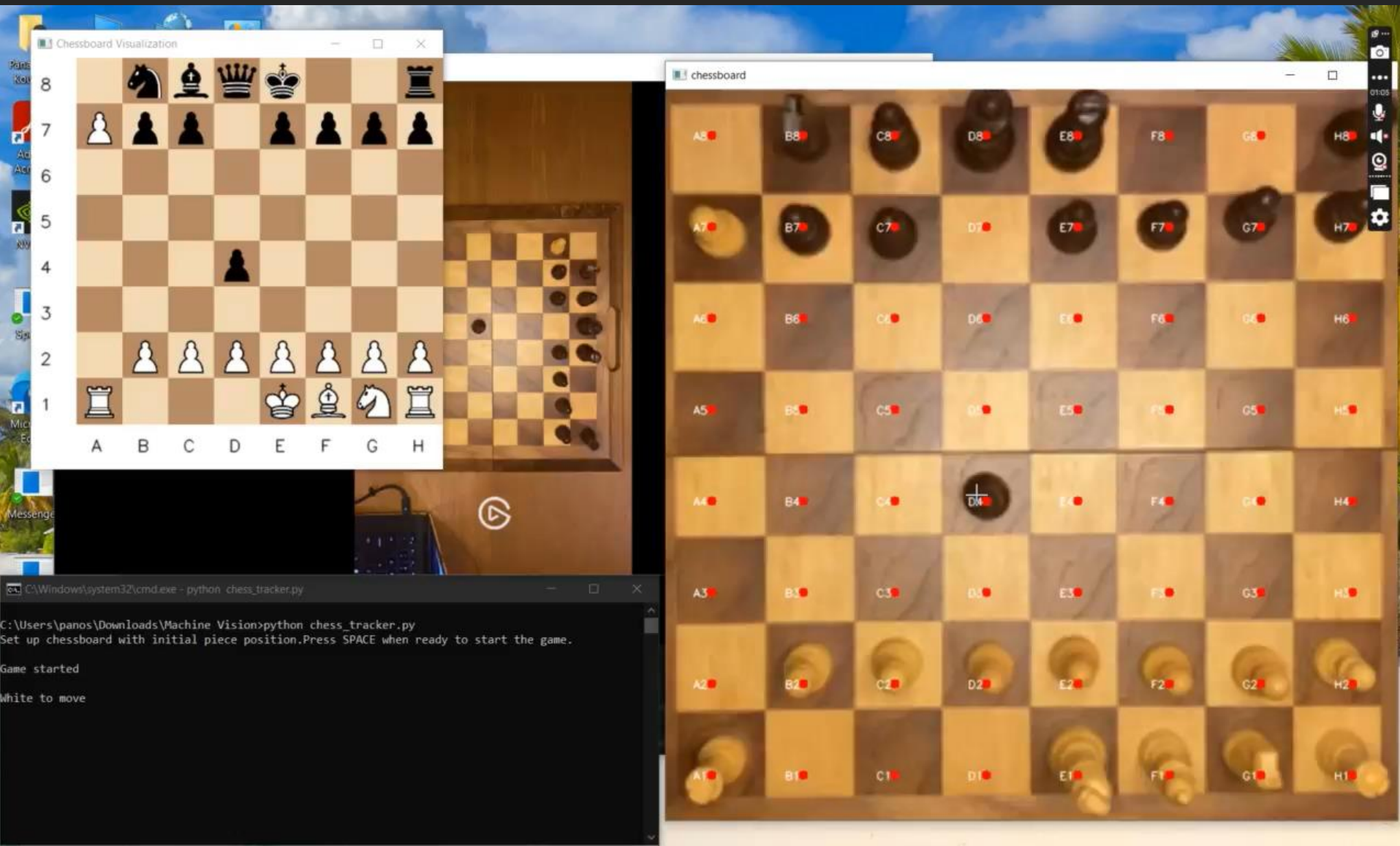


# A Different Outcome





# Special Moves



Thank you!

